The only parameters available at our disposal are the energy E and the undisturbed gas density ρ_0. The properties behind the shock wave such as p_1, ρ_1 are derivable from those in front of the shock wave. The only non-dimensional combination available from r, t, ρ_0 and E is

$$r\left(\frac{\rho_0}{Et^2}\right)^{1/5}.$$

It is reasonable to assume that the evolution in r and t of the shock wave depends only on the above variable. This means that the shock wave location $r = R(t)$ itself will correspond to a particular value, say β, of this variable, i.e.,

$$R = \beta\left(\frac{Et^2}{\rho_0}\right)^{1/5}.$$

The propagation velocity of the shock wave is

$$D = \frac{\mathrm{d}R}{\mathrm{d}t} = \frac{2R}{5t} = \frac{2\beta}{5}\left(\frac{E}{\rho_0 t^3}\right)^{1/5}$$

With the approximation described above, Rankine–Hugoniot conditions determines the gas velocity immediately behind the shock front v_1, p_1 and ρ_1 for an ideal gas as follows

$$v_1 = \frac{2}{\gamma+1}D, \quad p_1 = \frac{2}{\gamma+1}\rho_0 D^2, \quad \rho_1 = \rho_0\frac{\gamma+1}{\gamma-1}$$

where γ is the specific heat ratio. Since ρ_0 is a constant, the density immediately behind the shock wave is not changing with time, whereas v_1 and p_1 decrease as $t^{-3/5}$ and $t^{-6/5}$, respectively.